

YANAUL, Russian Federation

WMO: 284190

Lat: 56.2767N

Lon: 54.9690E

Elev: 99

StdP: 100.14

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-28.9	-34.7	0.1	-32.7	-31.2	0.2	-28.6	12.5	-6.5	12.1	-6.1	0.6	160	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	30.0	20.4	28.2	19.5	26.4	18.5	21.5	28.1	20.5	26.7	19.4	25.0	3.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	25.0	18.2	13.2	23.6	17.2	12.4	22.4	62.9	28.3	59.4	26.7	55.7	24.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.8	8.2	DB	-38.6	32.9	4.2	2.3	-41.7	34.5	-44.1	35.9	-46.5	37.2	-49.5	38.9	(4)
(5)			WB	-38.7	23.2	4.2	1.3	-41.7	24.2	-44.1	24.9	-46.5	25.7	-49.5	26.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-12.5	-12.6	-5.9	2.8	11.3	16.2	19.0	15.8	10.8	4.1	-6.1	-12.1
	DBStd	12.93	7.87	8.33	4.99	5.86	4.96	4.19	3.35	3.88	4.38	4.36	7.22	8.33
	HDD10.0	3501	698	632	493	224	46	3	0	2	43	191	482	687
	HDD18.3	5832	956	866	751	464	222	91	35	99	229	442	732	945
	CDD10.0	817	0	0	0	10	87	189	278	181	66	7	0	0
	CDD18.3	107	0	0	0	0	4	26	54	20	2	0	0	0
	CDH23.3	1238	0	0	0	2	79	299	595	226	37	0	0	0
	CDH26.7	346	0	0	0	0	9	87	171	71	8	0	0	0
Wind	WSAvg	3.3	3.9	4.0	4.0	3.2	3.3	2.7	2.0	2.3	2.7	3.7	3.6	4.0
Precipitation	PrecAvg	512	31	23	26	26	42	62	54	59	51	58	43	30
	PrecMax	689	56	39	52	60	85	119	113	128	124	98	71	54
	PrecMin	360	6	5	0	0	14	10	12	9	4	13	5	10
	PrecStd	72	13	10	13	17	21	29	28	26	27	22	18	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.9	4.3	22.8	28.2	32.9	31.8	32.0	27.6	19.0	6.1	1.8
		MCWB	0.7	0.6	2.0	13.6	15.9	20.4	21.1	19.9	16.3	11.4	5.1	0.8
	2%	DB	0.3	0.9	2.7	19.0	25.9	28.9	30.1	28.5	23.2	16.1	4.6	0.9
		MCWB	-0.4	0.2	1.2	11.0	15.8	19.8	20.8	19.6	15.2	10.2	3.8	0.2
	5%	DB	-0.9	0.1	1.8	15.6	23.5	26.7	28.7	25.7	20.6	13.1	3.2	0.0
		MCWB	-1.6	-0.5	0.7	9.8	14.9	19.0	20.4	18.3	14.4	9.5	2.3	-0.5
	10%	DB	-2.3	-1.2	1.0	12.1	21.1	24.5	27.0	23.0	17.8	10.7	1.8	-1.4
		MCWB	-2.9	-1.8	-0.1	7.9	13.7	17.9	19.6	17.1	13.1	8.6	1.1	-1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.0	2.5	13.9	18.2	22.2	22.7	22.3	18.6	12.9	5.3	1.1
		MCDB	0.9	1.4	3.4	22.0	24.8	29.2	29.1	28.9	23.7	15.8	5.6	1.4
	2%	WB	-0.3	0.3	1.5	11.8	16.7	20.8	21.8	20.2	16.1	11.3	3.9	0.4
		MCDB	0.2	0.7	2.4	17.5	23.4	27.4	28.4	26.3	21.0	15.1	4.4	0.9
	5%	WB	-1.5	-0.5	0.9	9.8	15.5	19.7	21.0	18.9	14.8	9.7	2.5	-0.5
		MCDB	-1.0	0.0	1.8	15.3	21.9	25.5	27.5	24.6	19.6	12.4	3.1	0.0
	10%	WB	-2.8	-1.8	0.1	7.9	14.2	18.5	20.0	17.7	13.7	8.4	1.2	-1.9
		MCDB	-2.3	-1.2	0.9	12.0	20.8	23.4	25.7	22.1	17.9	10.8	1.7	-1.5
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.4	8.4	9.5	11.0	11.1	11.5	10.1	9.6	6.0	5.8	7.1
		MCDBR	6.2	5.1	6.1	14.8	15.5	14.7	14.7	14.8	14.2	11.6	5.1	5.8
		MCWBR	6.2	4.9	5.1	8.2	7.6	6.9	6.4	6.5	7.0	6.8	4.7	5.7
	5% WB	MCDBR	6.3	5.3	5.2	13.9	14.1	13.1	13.7	13.0	12.2	9.9	4.8	5.7
		MCWBR	6.2	5.2	4.6	8.1	7.8	6.7	6.5	6.4	6.9	6.4	4.6	5.6
Clear-Sky Solar Irradiance	taub	0.299	0.322	0.377	0.410	0.399	0.394	0.414	0.435	0.381	0.358	0.317	0.295	
	taud	2.009	2.011	1.924	2.059	2.264	2.333	2.287	2.201	2.322	2.251	2.123	1.984	
	Ebn at Noon	546	701	750	797	840	849	820	769	763	669	530	436	
	Edh at Noon	66	99	139	142	123	117	120	123	95	79	59	53	
All-Sky Solar Radiation	RadAvg	0.69	1.66	3.06	4.29	5.37	5.71	5.71	4.38	2.70	1.18	0.53	0.39	
	RadStd	0.09	0.21	0.31	0.48	0.54	0.51	0.59	0.39	0.28	0.23	0.11	0.09	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page